

1) The specific heat of iron is 0.451 J/g K. What is the molar specific heat of iron?

A. 0.451 J/mol K B. 26.0 J/mol K

C. 25.2 J/mol K D. 55.85 J/mol K

$$\begin{aligned} C_m &= C_s + MM \\ &= 0.451 + 55.845 \\ &= 25.2 \text{ J/mol} \cdot \text{K} \end{aligned}$$

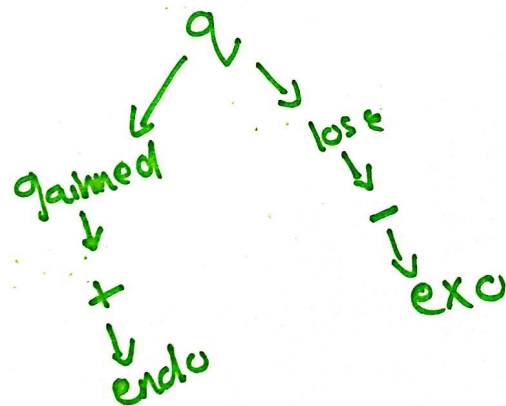
2) When heat is transferred to the system, the process is said to be endo and the sign of q is +.

A. exothermic, positive.

☒ B. endothermic, positive

C. exothermic, negative

D. endothermic, negative



3) What is ΔE for a system which has the following two steps?

Step 1: The system absorbs 60 J of heat while 40 J of work are performed on it.

Step 2: The system releases 30 J of heat while doing 70 J of work.

A. 110 J B. 100 J C. zero D. 90 J

$$\begin{aligned}\Delta E &= \underline{q} + \underline{w} \\ &= \frac{+60 + 40}{+100} - \frac{30 + 70}{-100} \\ &= \text{zero}\end{aligned}$$

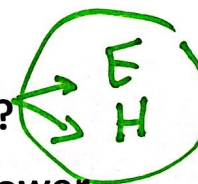
Which of the following is a state function?

A. enthalpy

B. work

C. heat

D. power



For a particular process $q = 30 \text{ kJ}$ and $w = -25 \text{ kJ}$. What conclusion may be drawn for the process?

→ gained endo

→ by the sys

$$\Delta E = 30 - 25 = 5 \text{ kJ}$$

A. $\Delta E = 55 \text{ kJ}$ X

B. the system does work on the surroundings. ✓

C. the heat transfer from the system to the surrounding. X

D. the work energy transfer from the surrounding to the system. X

How many joules of heat are required to heat 1.00 g of lead from 25 °C to the melting point (327. °C) and melt all of it? (The specific heat capacity of lead is 0.159 J/g K and it requires 24.7 J/g to convert lead from the solid to the liquid state.)

A. 2.47 J

B. 39.4 J

C. 48.0 J

D. 72.7 J

$$C_s = \frac{q}{m \cdot \Delta T}$$

$$q = 0.159 \times 1 \times (327 - 25) = 48.018$$

$$+ (24.7)$$

$$27.718 \text{ J}$$

~~$$27.718 \text{ J}$$~~

$$72.718 \text{ J}$$

When 221 grams of water at a temperature of 25.0 °C is mixed with an unknown mass of water at a temperature of 57.0 °C the final temperature of the resulting mixture is 33.1 °C. What was the mass of the second sample of water?

- A. 29.5 g B. 67.0 g C. 74.9 g D. 131 g

$$q = -q$$
$$\cancel{c} \cdot m \cdot \Delta T = -\cancel{c} \cdot m \cdot \Delta T$$
$$221 \times (33.1 - 25) = -m_2 \times (33.1 - 57)$$
$$m = 74.9 \text{ g}$$

When 108 grams of water at a temperature of 22.5 °C is mixed with 65.1 grams of water at an unknown temperature, the final temperature of the resulting mixture is 47.9 °C. What was the temperature of the other sample of water?

- A. 18.9 °C B. 67.0 °C C. 79.7 °C D. 90.0 °C

$$\begin{aligned} q_{\text{H}_2\text{O}} &= -q_{\text{H}_2\text{O}} \\ \cancel{C} \times m \times \Delta T &= -\cancel{C} \times m \times \Delta T \\ 108 (47.9 - 22.5) &= 65.1 \times (47.9 - T_i) \\ \underline{T_i = 90^\circ\text{C}} \end{aligned}$$

Calculate the amount of heat required to heat 357 grams of ethylene glycol (antifreeze) from 0 °C to 110 °C. The specific heat of ethylene glycol is 2.42 J/g K.

- A. 16.2 kJ B. 39.3 kJ C. 71.5 kJ D. 95.0 kJ

$$C_s = \frac{q}{m \cdot \Delta T}$$

$$q = 2.42 \times 357 \times (110 - 0)$$
$$= 95 \text{ kJ } \underline{\underline{D}}$$

10. Calculate the ^{C_s} specific heat of a certain unknown metal, if it requires 195 joules of heat to raise the temperature of 12.1 grams of this metal by 34.6 °C.
- q
 ΔT
- A. 0.451 J/g K B. 0.335 J/g K C. 0.714 J/g K D. 0.816 J/g K

$$C_s = \frac{q}{m \cdot \Delta T}$$

$$= \frac{195}{12.1 \cdot 34.6} = 0.466 \quad \underline{A}$$

11. If 0.750 g of MgO is placed in a coffee-cup calorimeter and then 100.0 mL of 1.00 M HCl is added, the temperature of the HCl solution increases from 22.8 °C to 28.7 °C. Based on this information, calculate the enthalpy change for this reaction per mol of MgO. You may assume that the specific heat of the solution is 4.20 J/g K and the density of the HCl solution is 1.00 g/mL.

- A. -2.54 kJ/mol B. -29.8 kJ/mol
C. -65.3 kJ/mol D. -134 kJ/mol

$$q_{\text{rxn}} = -C_s \times \text{total mass} \times \Delta T$$

$$\Delta H = \frac{q}{n}$$

$$q_{\text{rxn}} = -4.2 \times 0.75 + 100 \text{ mL} \times 1 \text{ g/mL} \times (-2.8)$$

$$= -2.5 \text{ kJ}$$

$$\Delta H = \frac{q}{n} = \frac{-2.5}{\left(\frac{0.75}{40.3}\right)} = \underline{\underline{-134 \text{ kJ/mol}}}$$



12. When a piece of unknown metal weighing 24.4 g, and at 119.3 oC, is placed in a calorimeter containing 51.0 g of water at 19.6 oC, the temperature increases to 20.8 oC. If the specific heat of the water is 4.18 J/g K and the specific heat of the calorimeter is 159 J/K, what is the specific heat of the unknown metal?

- A. 0.128 J/g K B. 0.186 J/g K C. 0.723 J/g K D. 1.12 J/g K

$$q = -q_v$$

$$C_s \cdot M \cdot \Delta T = - [C_{cal} \cdot \Delta T + C_{cal} \cdot \Delta T]$$

~~$$C_s \cdot 24.4 \cdot (20.8 - 119.3) = -1.2 [4.18 \cdot 51 + 159]$$~~

$$C_s \cdot 24.4 \cdot (20.8 - 119.3) = -1.2 [4.18 \cdot 51 + 159]$$

$$C_s = \underline{\underline{0.186}}$$

13. The standard molar enthalpy of combustion for $\text{C}_3\text{H}_8(\text{g})$ is -2044 kJ



What is the standard enthalpy change for the combustion of 3.000 mols of $\text{C}_3\text{H}_8(\text{g})$? $3 \text{C}_3\text{H}_8(\text{g}) + 15 \text{O}_2 \longrightarrow 9\text{CO}_2(\text{g}) + 12 \text{H}_2\text{O}(\text{l})$ ΔH_2

- A. -6132 kJ B. -4088 kJ C. -2044 kJ D. $+2044 \text{ kJ}$

$$\begin{aligned}\Delta H_2 &= 3 \times \Delta H_1 \\ &= 3 \times -2044 \text{ kJ} \\ &= -6132 \text{ kJ}\end{aligned}$$

14. What is the value of the molar ^{for 1 mole} enthalpy (or heat) of combustion of $\text{C}_2\text{H}_6(\text{g})$ if the combustion of 3.01 grams of this gas at constant pressure releases 8.47 kilojoules of heat?

- A. -0.847 kJ/mol B. -8.47 kJ/mol C. -84.7 kJ/mol D. -847 kJ/mol

$$\frac{3.01 \text{ g}}{30.07} \rightarrow -8.47 \text{ kJ}$$

1 mole ?? X

$$X = \frac{-8.47}{.01} = -84.7 \text{ kJ}$$

15. The molar enthalpy (or heat) of combustion of C₂H₅OH is -277.7 kJ/mol. How many kJ of heat will be released by the combustion of 9.22 g of C₂H₅OH at constant pressure?

- A. 11.8 kJ B. 27.8 kJ C. 32.9 kJ D. 55.5 kJ

$$n = \frac{9.22}{46.07} = \underline{\underline{0.2}}$$

$$\begin{array}{lcl} 1 & \longrightarrow & -277.7 \\ 0.2 & \longrightarrow & X \end{array}$$

$$X = -55.58 \text{ kJ}$$

16. Calculate the standard molar enthalpy of formation for Na₂O(s), given that the standard enthalpy of formation for Na₂O₂(s) is -505 kJ/mol and the enthalpy change for the following reaction

Na₂O(s) + 1/2 O₂ → Na₂O₂(s) is -89.0 kJ/mol.

A. 594 kJ/mol B. 416 kJ/mol C. -416 kJ/mol D. -594 kJ/mol

$$\begin{aligned}\Delta H_{\text{rxn}}^{\circ} &= \Delta H_f^{\circ} \text{ products} - \Delta H_f^{\circ} \text{ reactants} \\ -89 &= -505 - \left(\left(\frac{1}{2} \cdot 0 \right) + (X) \right) \\ -89 &= -505 - X \\ X &= -505 + 89 \\ &= -416 \text{ kJ/mol}\end{aligned}$$

Calculate the standard molar enthalpy of combustion for ethene, C₂H₄,



based on the following standard enthalpies of formation: $\Delta H^\circ_f[\text{C}_2\text{H}_4(\text{g})] = +52.26 \text{ kJ/mol}$; $\Delta H^\circ_f[\text{CO}_2(\text{g})] = -393.5 \text{ kJ/mol}$; and $\Delta H^\circ_f[\text{H}_2\text{O}(\text{g})] = -241.8 \text{ kJ/mol}$.

A. -204.3 kJ

B. -583.8 kJ

C. -687.6 kJ

D. -1322.9 kJ

$$\begin{aligned}\Delta H_{\text{rxn}}^\circ &= \Delta H_f^\circ (\text{products}) - \Delta H_f^\circ (\text{reactants}) \\ &= [(2 \times -241.8) + (2 \times -393.5)] - 52.26 \\ &= -1323.9 \text{ kJ}\end{aligned}$$

The standard molar enthalpy change is -802.3 kJ for the combustion of $\text{CH}_4(\text{g})$.



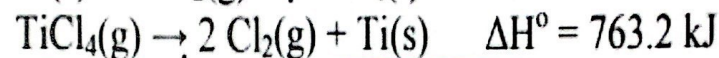
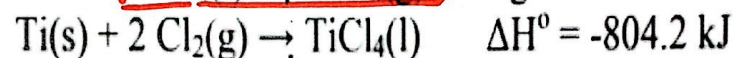
Calculate the standard molar enthalpy of formation for $\text{CH}_4(\text{g})$ based on the following standard enthalpies of formation: $\Delta H^\circ_f[\text{CO}_2(\text{g})] = -393.5 \text{ kJ/mol}$ and $\Delta H^\circ_f[\text{H}_2\text{O}(\text{g})] = -241.8 \text{ kJ/mol}$.

- A. -1679 kJ/mol B. -125.4 kJ/mol C. -74.8 kJ/mol D. +892.4 kJ/mol

$$\begin{aligned}\Delta H^\circ_{\text{rxn}} &= \Delta H^\circ_f \text{P} - \Delta H^\circ_f \text{R} \\ -802.3 &= [(2 \times -241.8) + (2 \times -393.5)] - \Delta H^\circ_f(\text{CH}_4) \\ \Delta H^\circ_f(\text{CH}_4) &= -74.8 \text{ kJ/mol}\end{aligned}$$

19. Calculate the enthalpy of vaporization for TiCl_4

$\text{TiCl}_4(\text{l}) \rightarrow \text{TiCl}_4(\text{g})$ given the following enthalpies of reaction:



A. -80.4 kJ

B. +41.0 kJ

C. +80.4 kJ

D. +127.3 kJ

Hess's Law

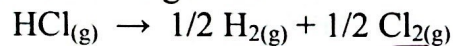
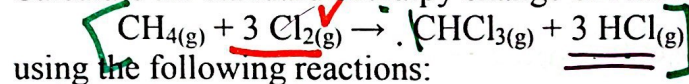


$$\Delta H = 804.2$$

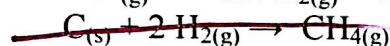
$$\Delta H = 763.2$$

$$+41 \text{ kJ}$$

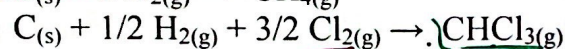
20. Calculate the standard enthalpy change of reaction for the process



$\Delta H^\circ = +92.3 \text{ kJ}$



$\Delta H^\circ = -74.8 \text{ kJ}$



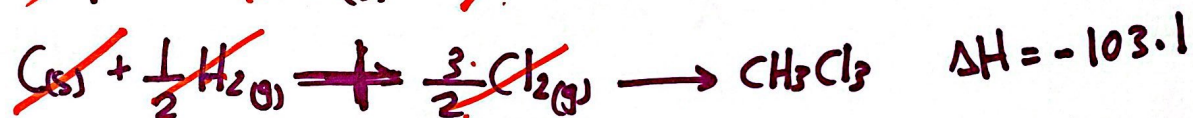
$\Delta H^\circ = -103.1 \text{ kJ}$

A. -120.6 kJ

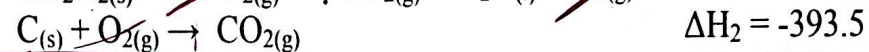
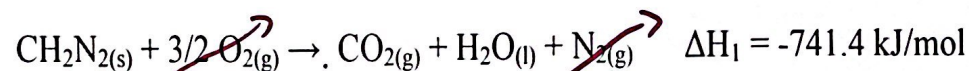
B. -212.9 kJ

C. -305.2 kJ

D. -454.8 kJ



21. Calculate the standard enthalpy of formation for CH_2N_2 , given the following standard enthalpies of reaction:



A. +62.1 kJ/mol

B. -633 kJ/mol

C. -849 kJ/mol

D. -1420 kJ/mol

$$\Delta H_{\text{rxn}} = \Delta H_{\text{f, P}} - \Delta H_{\text{f, R}}$$

$$-741.4 = (-285.8 + -393.5) - X$$

$$X = 62.1 \text{ kJ/mol}$$

✓